

240GS/s. This puts engineers and scientists working on fundamental scientific research, next-generation communication systems or high-bandwidth electrical components in the driver's seat.

Multi-function scopes for specific applications

It is now possible for systems using low-power double-data-rate 4 memory (LPDDR4) to be tested and characterised effectively and efficiently. With the help of an application that runs on Keysight's Infiniium 90000A, 90000 X-Series and Z-Series oscilloscopes, engineers can specifically test and validate LPDDR4 designs to ensure compliance with LPDDR4 JEDEC JESD209-1 standard.

Called Keysight N6462A DDR4 and LPDDR4 test application, it is designed for engineers who develop tablets, gaming consoles, smartphones and other consumer electronics. Keysight also claims that it is the only T&M company offering solutions across all stages of DDR chip development, be it hardware or software. It has also introduced measurement capabilities for applications making use of pulse-amplitude modulation with four amplitude level (PAM-4) signals.

In September 2014, Tektronix had also announced a test solution for LPDDR4, which is slated for adoption in 2015.

Tektronix recently introduced a set of USB 3.1 compliance test solutions, which allow for faster verification of designs against latest USB specifications. To get USB 3.1 certification, receiver testing is compulsory. Due to ever-rising data rates, it is important to make sure that the receiver correctly interprets the incoming bit stream. This solution, TekExpress USB 3.1, coupled with select Tektronix oscilloscopes can be used to debug and validate new USB power delivery ICs and system designs, and opens a broad range of applications for USBs.

Tektronix also launched a solution

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for engineers to test mobile industry processor interface (MIPI) M-PHY transmitters and receivers. MIPI M-PHY serial interface technology is gaining importance for next-generation mobile devices. This solution can only be used with Tektronix MSO/DPO 70000DX oscilloscopes and P7600 series probes.

Rohde & Schwarz, another T&M supplier, offers solutions for controller area network (CAN) flexible data (CAN FD) interface protocol in its RTE and RTO line of oscilloscopes. This would enable design engineers to analyse CAN interfaces, which support high-speed CAN FD bus protocol. CAN interfaces are increasingly being adopted in automotive and industry applications.

In the future

Vendors will continue to invest in the development of technologies with new application-specific integrated circuits (ASICs) and software platforms for future oscilloscopes.

Researchers at ARC Centre for ultra-high bandwidth devices for optical systems (CUDOS) at University of Sydney and Australian National University have developed an optical oscilloscope that boasts of 20 times the resolution of conventional electronics. As a result, we could possibly see commercialisation of this next generation of photonic technology developed by researchers.

On University of Sydney's website, Dr Simon Lefrancois, from University of Sydney's School of Physics, was quoted as saying, "This is the



Dr Simon Lefrancois from University of Sydney's School of Physics holding an optical chip used in the terabit oscilloscope (Image courtesy: <http://sydney.edu.au>)

culmination of four years of research and discussion with industry partners. The ultra-high resolution we have achieved is made possible by directly analysing optical data with a laser beam rather than converting it to a slower electric signal. Our specialty glass greatly enhances sensitivity of this interaction so that it can be done with low power on a compact optical chip."

In an article by Chris Lee, a physicist and science writer, he shares that a group of engineers have made an oscilloscope based on laser-triggered field emitters. By this, an oscilloscope can display changes in voltage and current at very high speeds. This is still in an experimental phase and although incorporating laser throws a variety of challenges, it is still achievable. These oscilloscopes would likely fall under the high-end range for research and industrial purposes. ●